Stability

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Multivariate Stability measures for each individual

Figure 4C in publication

```
df <- MEconc_newlong %>%
  ungroup %>%
  filter(!is.na(`Sample ID`)) %>%
  filter(!str_detect(met,'PC')) %>%
  filter(!(str_detect(met,'glycerol') | met == 'ERECTIC' | met=='TSP' | str_detect(met,'PC'))
  filter(!(str_detect(met,'glycerol') | str_detect(met,'EDTA') | met == 'ERECTIC' | met=='TSP
  dplyr::select(`Sample ID`, met,logy) %>%
  spread(met,logy)
DMENMR <- dist(scale(df[,-1]))
medianNMR <- DMENMR %>% as.vector() %>% median()
df <- df %>% mutate(Participant = `Sample ID` %>% substr(1,1))

phyXsel16s <- phyX %>%
  subset_samples(!is.na(Sample.ID)) %>%
  transform_sample_counts( function(x) x  / sum(x))
D16s <- phyXsel16s %>% phyloseq::distance("bray")
median16s <- D16s %>% as.vector() %>% median()

phyXselits <- phyITS %>%
```

```
  subset_samples(!is.na(Sample.ID)) %>%
  transform_sample_counts( function(x) as.numeric(x>0))
DITS <- phyXselits %>% phyloseq::distance("euclidean")
medianITS <- DITS %>% as.vector() %>% median()


res_withindist <- data.frame()
for (P in unique(df$Participant)){
  ic <- df$Participant==P
  # print(sum(ic))
  #print(sum(ic))
  iD <- as.matrix(DMENMR)[ic,ic] / medianNMR # scale distance to equal median.
  res_withindist <- res_withindist %>%
    bind_rows(data.frame(Participant = P,
                         nsamples = sum(ic),
                         withinparticipantdist = iD[upper.tri(iD)],
                         omics = 'ME-NMR'))

  ic <- phyXsel16s@sam_data$Participant==P
  iD<- as.matrix(D16s)[ic,ic] / median16s

  if (length(iD)==1) {wd <- NA} else {wd <- iD[upper.tri(iD)]}
  res_withindist <- res_withindist %>%
    bind_rows(data.frame(Participant = P,
                         nsamples = sum(ic),
                         withinparticipantdist = wd,
                         omics = 'ME-16s'))

  ic <- phyXselits@sam_data$Participant==P
  iD<- as.matrix(DITS)[ic,ic] / medianITS
  if (length(iD)==1) {wd <- NA} else {wd <- iD[upper.tri(iD)]}

  res_withindist <- res_withindist %>%
    bind_rows(data.frame(Participant = P,
                         nsamples = sum(ic),
                         withinparticipantdist = wd,
                         omics = 'ME-ITS'))
}

res_withindistsum <- res_withindist %>%
  group_by(Participant,omics) %>%
  summarise(n_comparisons = n(), mnd = mean(withinparticipantdist),
```

```
            nsamples = mean(nsamples))

res_withindistsum_sens_8 <- res_withindist %>%
  group_by(Participant,omics) %>%
  mutate(n_comparisons = n(),
            nsamples = mean(nsamples)) %>%
  filter(nsamples>7) %>%
  group_by(Participant,omics) %>%
  summarise(mnd = mean(sample(withinparticipantdist)[1:8])) %>%
  mutate(type = '8 samples per indiv')

res_withindistsum_sens_4 <- res_withindist %>%
  group_by(Participant,omics) %>%
  mutate(n_comparisons = n(),
            nsamples = mean(nsamples)) %>%
  filter(nsamples>3) %>%
  group_by(Participant,omics) %>%
  summarise(mnd = mean(sample(withinparticipantdist)[1:4])) %>%
  mutate(type = '4 samples per indiv')

res_withindistsum %>% group_by(omics) %>%
  summarise(sum(nsamples),
            length(unique(Participant)),
            quantile(nsamples,0.75),
            median(nsamples),
            quantile(nsamples,0.25)) %>%
  reactable() %>%
  add_subtitle('Descriptives of samples going into stability calculations')
```

Descriptives of samples going into stability calculations

omics	sum(nsamples)	length(unique(Participant))	quantile(nsamples, 0.75)	median(nsamples)	quantile(nsamples, 0.25)
ME-16s	141	18	9	9	8
ME-ITS	141	18	9	9	8
ME-NMR	110	18	8	6	5

```
r <- res_withindistsum %>%
  dplyr::select(-n_comparisons) %>%
```

```
  spread(omics,mnd) %>%
  ungroup %>%
  dplyr::select(-Participant) %>%
  rstatix::cor_mat()

pv <- res_withindistsum %>%
  dplyr::select(-n_comparisons) %>%
  spread(omics,mnd) %>%
  ungroup %>%
  dplyr::select(-Participant) %>%
  rstatix::cor_pmat()

FIG3b <- r %>% gather(omics,val,-rowname) %>%
  filter(val<1) %>%
  ggplot(data = ., aes(omics,rowname,fill = val, label= paste('r =',val))) +
  scale_fill_gradient2(low = 'blue',high = 'red', mid = 'white', midpoint = 0, limits = c(0,1
  geom_tile() + geom_text() +
  ylab(NULL) + theme_bw() +
  theme(legend.position = 'none')

FIG3a <- res_withindistsum %>%
  mutate(omics = factor(omics, levels = c('ME-NMR','ME-16s','ME-ITS'))) %>%
  ggplot(data = ., aes(omics, mnd, fill = omics)) +
  geom_violin() +
  geom_hline(yintercept = 1) +
  scale_fill_manual(values = colorcodes, breaks = colorcodesnames) +
  geom_line(aes(group = Participant)) +
  geom_point()+
  theme_bw() +
  ylab('Mean within participant distance') +
  theme(legend.position = 'none')

r %>% reactable() %>% add_subtitle('Correlation between overall stability')
```

Correlation between overall stability

rowname	nsamples	ME-16s	ME-ITS	ME-NMR
nsamples	1	0.13	0.064	-0.19
ME-16s	0.13	1	0.18	0.57
ME-ITS	0.064	0.18	1	0.075
ME-NMR	-0.19	0.57	0.075	1

```
pv %>% reactable() %>% add_subtitle('Inference (pv) between overall stability')
```

Inference (pv) between overall stability

rowname	nsamples	ME-16s	ME-ITS	ME-NMR
nsamples	0	0.607476357374591	0.800302412802145	0.4550494298685
ME-16s	0.607476357374591	1.16043393370025e-126	0.475433329431351	0.426941391723264
ME-ITS	0.800302412802145	0.475433329431351	2.97071087027252e-124	0.925397694650614
ME-NMR	0.4550494298685	0.426941391723264	0.925397694650614	0

Compare within-women stability with demographics

```
variablemetavars <- c("Alcohol","Coitus", "Medication during sampling"  )
metavars <- c("Age", "Residence", "Diet", "Contraceptives", "Regular/daily medication" , "Ni

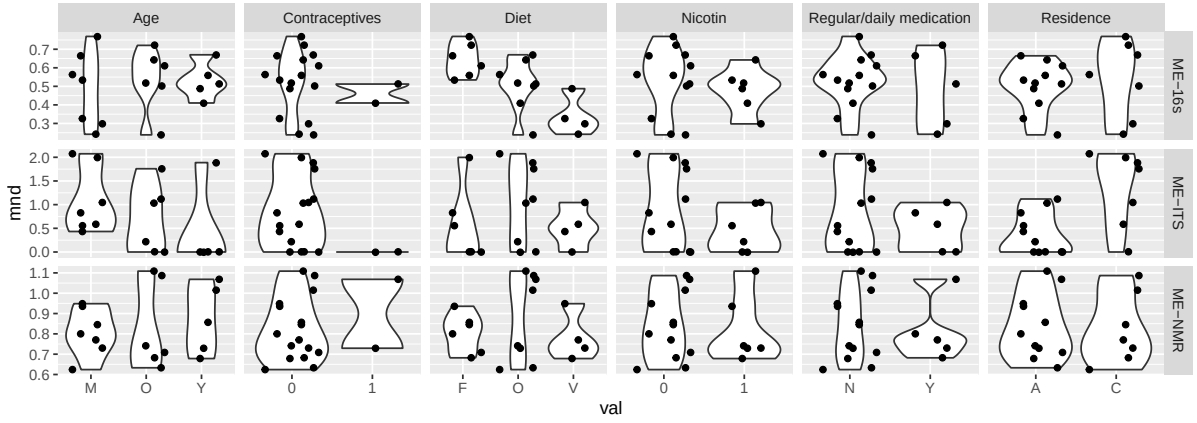
dfmeta <- MEconc_newlong %>%
  filter(!duplicated(`Samples #`)) %>%
  dplyr::select(`Samples #`,Participant,   metavars) %>%
  gather(demo,val,-`Samples #`,-Participant) %>%
  filter(!is.na(Participant)) %>%
  dplyr::select(-`Samples #`) %>%
  group_by(demo) %>%
  filter(!duplicated(Participant)) %>%
  full_join(res_withindistsum, by = 'Participant')
```

```
dfmeta_sens <- MEconc_newlong %>%
  filter(!duplicated(`Samples #`)) %>%
  dplyr::select(`Samples #`,Participant,   metavars) %>%
  gather(demo,val,-`Samples #`,-Participant) %>%
  filter(!is.na(Participant)) %>%
  dplyr::select(-`Samples #`) %>%
  group_by(demo) %>%
  filter(!duplicated(Participant)) %>%
  full_join(bind_rows(res_withindistsum_sens_8,res_withindistsum_sens_4), by = 'Participant')
  filter(!is.na(mnd))
```

```
fmrt <- list(
  relssq = colDef(format = colFormat(percent = TRUE, digits = 1)),

  p.value = colDef(
    cell = function(value) {
      if (is.na(value)) return("")
      formatC(value, format = "e", digits = 2)
    }
  )
)


dfmeta %>%
  ggplot(data = ., aes(val,mnd)) +
  geom_violin() +
  geom_jitter() +
  facet_grid(omics~demo, scales = 'free')
```



Correlation table

Inference as table

```
tb_sens <- dfmeta_sens %>%
  group_by(omics,demo,type) %>%
  do(lm(data = ., mnd~val) %>% anova %>% tidy) %>%
  group_by(demo,omics,type) %>%
  mutate(relssq = sumsq / sum(sumsq), nindiv = sum(df)+1) %>%
  filter(term=='val') %>%
  mutate(R2_pv = paste('(n = ',nindiv,' R2 = ',round(100*relssq,1),'%, p = ',round(p.value,3)
  dplyr::select(omics,demo,type,R2_pv) %>%
  spread(type,R2_pv)

dfmeta %>%
  ungroup %>%
  group_by(omics,demo) %>%
  do(lm(data = ., mnd~val) %>% anova %>% tidy) %>%
  group_by(demo,omics) %>%
  mutate(relssq = sumsq / sum(sumsq)) %>%
  filter(term=='val') %>%
  arrange(p.value) %>%
  dplyr::select(omics,demo,df,relssq,p.value) %>%
  left_join(tb_sens, by = c('omics','demo')) %>%
  reactable(columns = fmrt) %>%
    add_subtitle('Variation in stability explained by demographics')
```

Variation in stability explained by demographics

omics	demo	df	relssq	p.value	4 samples per indiv	8 san per i
ME-16s	Diet	2	52.1%	3.99e-03	(n = 17 R2 = 28.5%, p = 0.095)	(n = 54%, 0.014
ME-ITS	Residence	1	40.0%	4.89e-03	(n = 17 R2 = 31.7%, p = 0.019)	(n = 28.9% 0.047
ME-ITS	Contraceptives	1	13.1%	1.41e-01	(n = 17 R2 = 6.5%, p = 0.322)	(n = 7.3% 0.352
ME-ITS	Nicotin	1	7.0%	2.88e-01	(n = 17 R2 = 3.7%, p = 0.46)	(n = 10.4% 0.26)
ME-ITS	Age	2	14.9%	2.98e-01	(n = 17 R2 = 7.6%, p = 0.574)	(n = 9.2% 0.589
ME-ITS	Regular/daily medication	1	4.8%	3.80e-01	(n = 17 R2 = 3.3%, p = 0.487)	(n = 6.5% 0.377
ME-ITS	Diet	2	10.0%	4.54e-01	(n = 17 R2 = 21.2%, p = 0.189)	(n = 31.4% 0.126
ME-NMR	Contraceptives	1	2.4%	5.42e-01	(n = 15 R2 = 10.8%, p = 0.232)	(n = 93.9% 0.001
ME-16s	Nicotin	1	2.3%	5.48e-01	(n = 17 R2 = 4.7%, p = 0.401)	(n = 6.5% 0.38)
ME-NMR	Diet	2	6.9%	5.85e-01	(n = 15 R2 = 4.1%, p = 0.778)	(n = 10.2% 0.851

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```
dfmeta %>% filter(demo=='Diet') %>%
  mutate(Diet = val) %>%
  dplyr::select(omics,Participant,Diet,mnd) %>%
  rio::export('Figure5_data.xlsx')

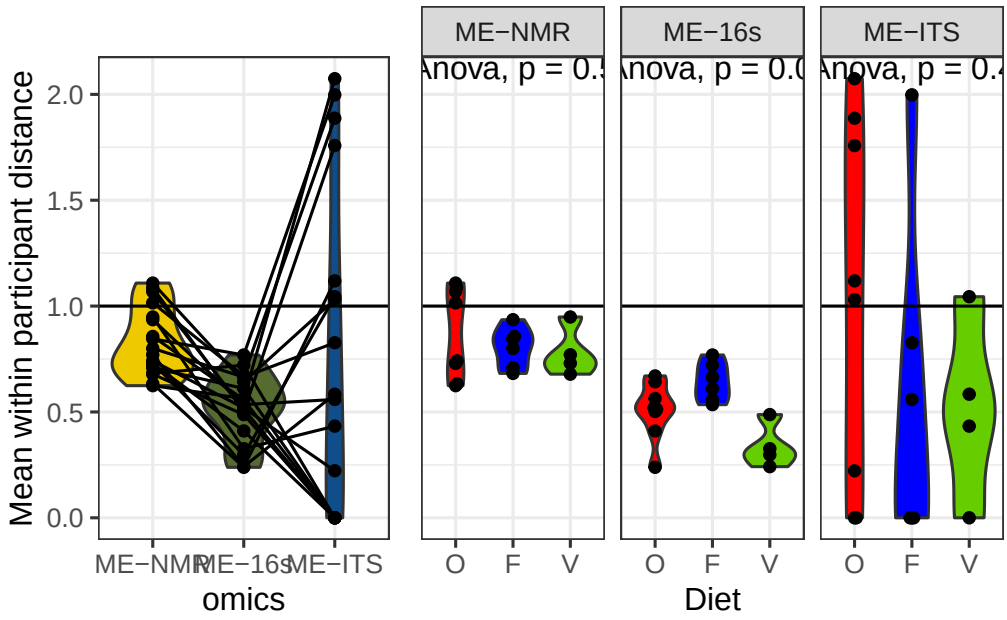
FIG3c <- dfmeta %>%
  filter(demo=='Diet') %>%
  mutate(omics = factor(omics, levels = c('ME-NMR','ME-16s','ME-ITS'))) %>%
  mutate(val = factor(val,levels = c('O','F','V'))) %>%
  ggplot(data = ., aes(val,mnd, fill = val)) +
  geom_violin() +
  geom_hline(yintercept = 1) +
  scale_fill_manual(values = c('red','blue','chartreuse3')) +
  ggbeeswarm::geom_beeswarm() +
  theme_bw() + theme(legend.position = 'none') +
  ylab('Mean within participant distance') + xlab('Diet') +
  ggpubr::stat_compare_means(method = 'anova') +
  facet_grid(.~omics)
```

Overall stability per women - across omics modalities

Figure 5 in publication Example on diet.

```
ggsave(filename = 'FIG5_betastab.pdf', plot = FIG3a, height = 5,width = 3)

layout <- "
ABB
"
FIG5 <- FIG3a + FIG3c + plot_layout(design = layout , axes = 'collect')
FIG5
```




```
ggsave(filename = 'FIG5_betastab.pdf', plot = FIG5, height = 5,width = 8)
```